ISHAQ PAKTIN YAR

Data Scientist · Backend Engineer · ML Researcher

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ML Engineer with a track record of delivering production-grade systems and measurable research impact — 18% forecasting lift, 98% protein classification accuracy, 60% reduction in admin overhead. Unique combination of deep learning research (CU Anschutz), full-stack production engineering (Bitatlım, Veroz), and academic excellence (B.Sc. High Honors, 2025). Fluent in 5 languages across 4 language families.

EDUCATION

B.Sc. Computer Engineering — High Honors
2021 – 2025

Istanbul Arel University

Thesis: "ClimatePulse: Multivariate Time-Series Forecasting for Air Quality & Climate" — Achieved 18% MAE improvement over ARIMA and LSTM baselines using multi-horizon deep learning (NeuralForecast + PyTorch).

Supervisor: Asst. Prof. Mert Yağcıoğlu

WORK EXPERIENCE

Software Engineer
Aug 2025 – Present

Parem Academy · *Istanbul (Freelance / Independent)*

- Architected and deployed 5+ production systems including Bitatlım (marketplace) and Veroz (logistics) as systems serving real users.
- Designed high-concurrency REST APIs and real-time tracking pipelines using Node.js and PostgreSQL.
- Led infrastructure decisions (self-hosted VPS, caching, CI/CD), cutting hosting costs by 90%.
- Mentored junior interns in backend architecture and production workflows.

Undergraduate Researcher
Jul 2024 – Present

JRAVI Lab — CU Anschutz · *Remote*

- Improved protein function prediction accuracy from 86% to 98% using classical ML and deep models.
- Benchmarked lightweight models against protein language models, achieving comparable accuracy at significantly lower compute cost.
- Developing species classification models for protein sequence analysis.
- Contributing to manuscript in preparation for peer-reviewed submission (JRAVI Lab, CU Anschutz).

IT Coordinator
Sep 2023 – Sep 2025

Istanbul&I · *Istanbul (Part-Time)*

- Designed and deployed a Django-based management system serving 1,800+ users, replacing fragmented Excel workflows.
- Built secure REST APIs for member analytics and operational tracking.
- Reduced administrative processing time by 60% through centralized relational database architecture.

Software Developer Intern
Feb 2025 – Aug 2025

DevSecure · *Istanbul*

- Maintained and debugged client web platforms across the full stack, resolving backend and database issues within active production environments.
- Onboarded and mentored 2–3 junior interns on codebase conventions, Git workflows, and deployment standards, reducing their ramp-up time.

Science & Programming Educator
Dec 2023 – Apr 2024

Bilim Üsküdar Science Center · *Istanbul (Part-Time)*

- Taught applied science and programming to 20+ students daily across rotating school groups, translating complex technical concepts into hands-on, age-appropriate sessions.
- Designed and delivered an original lesson demystifying neural networks; explaining backpropagation as scaled mathematics rather than "magic AI", making the concept accessible to young learners.

KEY PROJECTS

Bitatlım — Hyperlocal Marketplace & Logistics Platform *Next.js · NestJS · PostgreSQL · Redis · Linux VPS*

- Architected multi-tenant marketplace with hybrid logistics engine (Veroz); supports real-time courier pooling.
- Built cryptographic ledger for automated multi-party payouts and dual-layer API security (JWT rotation + HMAC-SHA256).
- Reduced hosting costs by 90% via optimized VPS infrastructure, Redis caching, and automated CI/CD.

Pathogenic Feature Discovery — Deep Sequence Modeling *Python · PyTorch · SHAP*

- Built Conv1D + Attention model for pathogenicity prediction from raw protein sequences.
- Engineered memory-efficient 8,928-dimension k-mer encoding pipeline with taxa-aware GroupKFold to prevent biological data leakage.
- Applied SHAP-based auditing to validate motif learning and rule out taxonomic bias.

Air Quality Multivariate Forecasting (Thesis) *Python · PyTorch · NeuralForecast*

- Developed 7-day multi-horizon forecasts for PM10, NO2, SO2, O3 using 54 features to model cross-pollutant temporal dependencies.
- Improved MAE by 18% over ARIMA and LSTM baselines; applied rolling-origin evaluation to address seasonality and concept drift.
- Engineered a custom RASA NLU chatbot enabling natural-language weather queries with real-time forecast integration.

Additional projects available at Portfolio

TECHNICAL SKILLS

Languages	**Python, TypeScript, SQL, Bash, Java, C++**
ML / AI	PyTorch, Transformers, CNNs · RNNs · LSTMs, Time-Series Forecasting, SHAP
Backend	Node.js (NestJS / Express), Django, Flask, REST APIs, ETL Pipelines
Data	PostgreSQL, Redis, Relational Modeling, Real-Time Pipelines
Infrastructure	Docker, Linux, AWS, Git, CI/CD, Self-Hosted VPS

LANGUAGES

Pashto (Native) · English C1 · Persian C1
Urdu C1 · Turkish B2 · Spanish A2

CERTIFICATIONS & COMMUNITY

IBM AI Engineering — IBM Coursera (2023)
IELTS — British Council (2024)
Founder, i&i Book Club — 90+ members (2024)
Data Commission Volunteer, T3 AI'LE (2024)